

Bidirectional translocation of actomyosin drives epithelial invagination in ascidian siphon tube morphogenesis

Reviewed Preprint

v1 • November 19, 2025

Not revised

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eLife Assessment

This **useful** study uses a combination of experimental and modeling approaches to investigate the role of actomyosin in epithelial invagination during *Ciona* siphon tube morphogenesis. Several types of **solid** quantitative analyses are presented, yet the evidence supporting the central claim of bidirectional translocation of actomyosin remains **incomplete**. Since epithelial invagination contributes to the morphogenesis of many developing organs, this work has the potential to appeal to both cell biologists and developmental biologists.

<https://doi.org/10.7554/eLife.108588.1.sa4>

Abstract

How epithelia preform a spatiotemporal heterogeneous force generating program to drive a sequential tissue morphogenesis remains unclear, particularly the underlying precise mechanical mechanisms. This study investigated dynamic actomyosin reorganization between apical and lateral membrane cortex regions during two sequentially invaginated stages during ascidian atrial siphon tube morphogenesis. At the initial invagination stage, the originally lateral-located actomyosin translocated to the apical domains, while those actomyosin re-translocated back to lateral domains at the accelerated invagination stage. Using genetic mutants to modulate myosin activities, the initial invagination was strengthened or abolished, indicating invagination are apical constriction dependent. Optogenetic inhibition of myosin activities in lateral domains after initial invagination stage blocked the further processes, suggesting lateral constriction of actomyosin is required for the accelerated invagination. Vertex model simulations uncovered a coupled mechanism underlying epithelial invagination driven by apicobasal tension imbalance and lateral contraction. We thus propose an actomyosin translocation mechanical model: lateral actomyosin first translocate apically to drive apical constriction and shape the initial invagination, then apical actomyosin redistributes laterally to promote lateral contractility

and accelerate invagination. Our findings discovery a bidirectional reorganization of actomyosin network as a central mechanism driving epithelial invagination, providing insights on epithelial invagination and the organ morphogenesis during development.

Introduction

Morphogenesis, by which cells and tissues acquire their shape and structure, is a central theme in developmental biology. The tissue shape transformation in morphogenesis is fundamentally driven by the spatiotemporally regulated biomechanical forces (Collinet & Lecuit, 2021 [↗](#); Matter & Balda, 2003 [↗](#); Štorgel et al., 2016 [↗](#)). The generation and conduction of biomechanical forces play as an essential mechanism of morphogenesis, via regulating cell rearrangement, division, and deformation at both cellular and tissue scales (Chan et al., 2017 [↗](#); Lecuit et al., 2011 [↗](#); Matter & Balda, 2003 [↗](#); Nagafuchi, 2001 [↗](#); Osswald et al., 2022 [↗](#); Tepass, 2002 [↗](#)). Among these, actomyosin contractility plays a pivotal structure generating biomechanical tension in subcellular level, which is the basic of cell and tissue-level morphogenesis (Chung et al., 2017 [↗](#); Even-Ram et al., 2007 [↗](#); Fristrom, 1988 [↗](#); Lee & Harland, 2007 [↗](#); Popkova et al., 2024 [↗](#); Vicente-Manzanares et al., 2009 [↗](#); Zartman & Shvartsman, 2010 [↗](#)). Despite the widely acknowledged critical role of bio-mechanical forces in morphogenesis, the precise mechanisms by which these forces regulate cell and tissue remodeling, particularly how cells reorganize contractile domains in space and time to coordinate sequential morphogenetic events remains a significant challenge.

A typical example of morphogenesis is the transition from simple epithelial sheets to complex three-dimensional architectures (Collinet & Lecuit, 2021 [↗](#)). A large number of previous researches demonstrated the function of apical constriction in epithelial sheet invagination, such as in the invagination process of the *Drosophila* embryonic trachea (Schottenfeld et al., 2010 [↗](#)) and salivary gland placode (Pearl et al., 2017 [↗](#)). However, more and more studies noticed that the apical constriction itself is not sufficient in shaping complex epithelial three-dimensional structures and various epithelial tube shapes (Kondo & Hayashi, 2013 [↗](#); Pearl et al., 2017 [↗](#)). So, an interesting question is how do epithelial cells preform a spatiotemporal heterogeneous force generating program to drive a sequential epithelial tissue reshaping.

Marine ascidian *Ciona* is an excellent model for studying morphogenesis due to its simple developmental process, transparent embryos, and its close evolutionary relationship with vertebrates (Blair & Hedges, 2005 [↗](#); Delsuc et al., 2006 [↗](#); Zhao et al., 2021 [↗](#)). For example, asymmetrical actomyosin contractility in notochord was demonstrated to provide force for tail bending (Lu et al., 2020 [↗](#); Peng et al., 2020 [↗](#)). Actomyosin sequentially localized on the apical and basolateral cell surfaces to drive the endodermal invagination (Fiuza & Lemaire, 2021 [↗](#); Hotta et al., 2007 [↗](#); Kourakis et al., 2010 [↗](#)). Local myosin activation coupled with junctional rearrangements drives directional zippering (Hashimoto & Munro, 2019 [↗](#); Hashimoto et al., 2015 [↗](#)) in neural tube closure, and cell cortex distribution and the stability of tight junctions (TJs) were essential for notochord tube lumen opening and expansion (Shi et al., 2025 [↗](#)).

Ciona and vertebrates share evolutionary developmental homology. There may be a potential homology between the otic placode of vertebrates and the atrial siphon primordium of *Ciona* (Kourakis et al., 2010 [↗](#)). The atrial siphon forms from a non-dividing region in the lateral-dorsal epidermis of the head and follows a separate developmental trajectory (Hotta et al., 2020 [↗](#); Kourakis et al., 2010 [↗](#)). Invagination initially generates shallow pits, which later deepen and connect to the gut lumen after metamorphosis, creating two atrial siphons (Hotta et al., 2007 [↗](#); Kourakis et al., 2010 [↗](#)). Eventually, the left and right siphons fuse to form a single atrial siphon (Chiba et al., 2004 [↗](#)). Despite these observations, the early cellular events during atrial siphon tube morphogenesis remain incompletely understood.

In this study, we investigated the morphogenesis of the *Ciona* atrial siphon. Firstly, the detailed cellular process of *Ciona* atrial siphon formation through initial and the accelerated stages was described by visualizing cell boundary and nuclei. By combining actomyosin localization analysis, genetic manipulation, and vertex model simulations, we demonstrated that lateral actomyosin first translocated apically to drive apical constriction and established the initial invagination, then redistributed laterally to promote cell shortening and the further atrial siphon tube invagination. Disruption of myosin activity altered invagination timing, while optogenetic inhibition of myosin activity after initial apical constriction stage blocked the further invagination. Furthermore, a cell-based vertex model was established to validate the relationship between contractile force redistribution and epithelial invagination. The coupled biomechanical mechanisms induced by apicobasal imbalance and lateral contraction were uncovered as fundamental determinants of the atrial siphon invagination. These results provide new insights into the biomechanical control of tissue morphogenesis, and highlight the value of marine model organisms for addressing fundamental questions in developmental biology.

Results

The cellular processes of *Ciona* atrial siphon invagination

To investigate the early cellular events during atrial siphon morphogenesis, we performed actin filament (F-actin) staining on the fixed samples to capture the detailed cellular dynamics (**Figure 1A**). The cell at the bottom center of the invaginating region was defined as the “center cell” (**Figure 1B**). Measurements taken at different stages included the height and apical-to-basal area ratio of the center cell, the invagination depth of the atrial siphon and the linear distance between the $-3/-4$ and $+3/+4$ cell junctions (**Figure 1B-E**, **Figure 1**—figure supplement 1). At the initial invagination stage (13.5–16.0 hpf), the center cell height increased, the apical-to-basal area ratio decreased, and the protruding cell surface became inward pit, but no significant invagination occurred (**Figure 1A**), as reflected by a low invagination slope ($k = 0.2617$). During the accelerated stage (16.0–18.0 hpf), the center cell height decreased, the apical-to-basal area ratio stabilized, and invagination progressed rapidly ($k = 2.7920$). EdU incorporation and TUNEL assays demonstrated that neither cell division nor apoptosis was involved in the whole invagination process (**Figure 1**—figure supplement 2).

Bidirectional translocation of actomyosin between apical and lateral domains during atrial siphon tube invagination

To analyze the forces driving atrial siphon tube invagination, we first visualized the distribution of F-actin and quantified its dynamics at different cellular domains (**Figure 1**—figure supplement 1A). During the initial invagination stage, before 15 hpf, F-actin concentration decreased at the lateral domains, while gradually increased around the apical membrane, resulting in a higher F-actin accumulation at the apical region. Subsequently, the F-actin concentration at lateral region started to increase and exceeded that in the apical domain after 16 hpf, during which apical F-actin levels showed a gradual decline (**Figure 1**—figure supplement 1A).

Using anti-pS19 MRLC antibody, we examined the spatial and temporal patterns of myosin activity within the atrial siphon primordium. Quantification of signal intensities from the apical, basal, and lateral regions during the initial and the accelerated stages showed that the distribution pattern of active myosin was consistent with that of F-actin (**Figure 2A, B**). Specifically, active myosin and F-actin levels were higher at the apical region than the lateral one during the initial stage, whereas the opposite pattern was observed during the accelerated stage (**Figure 2B**).

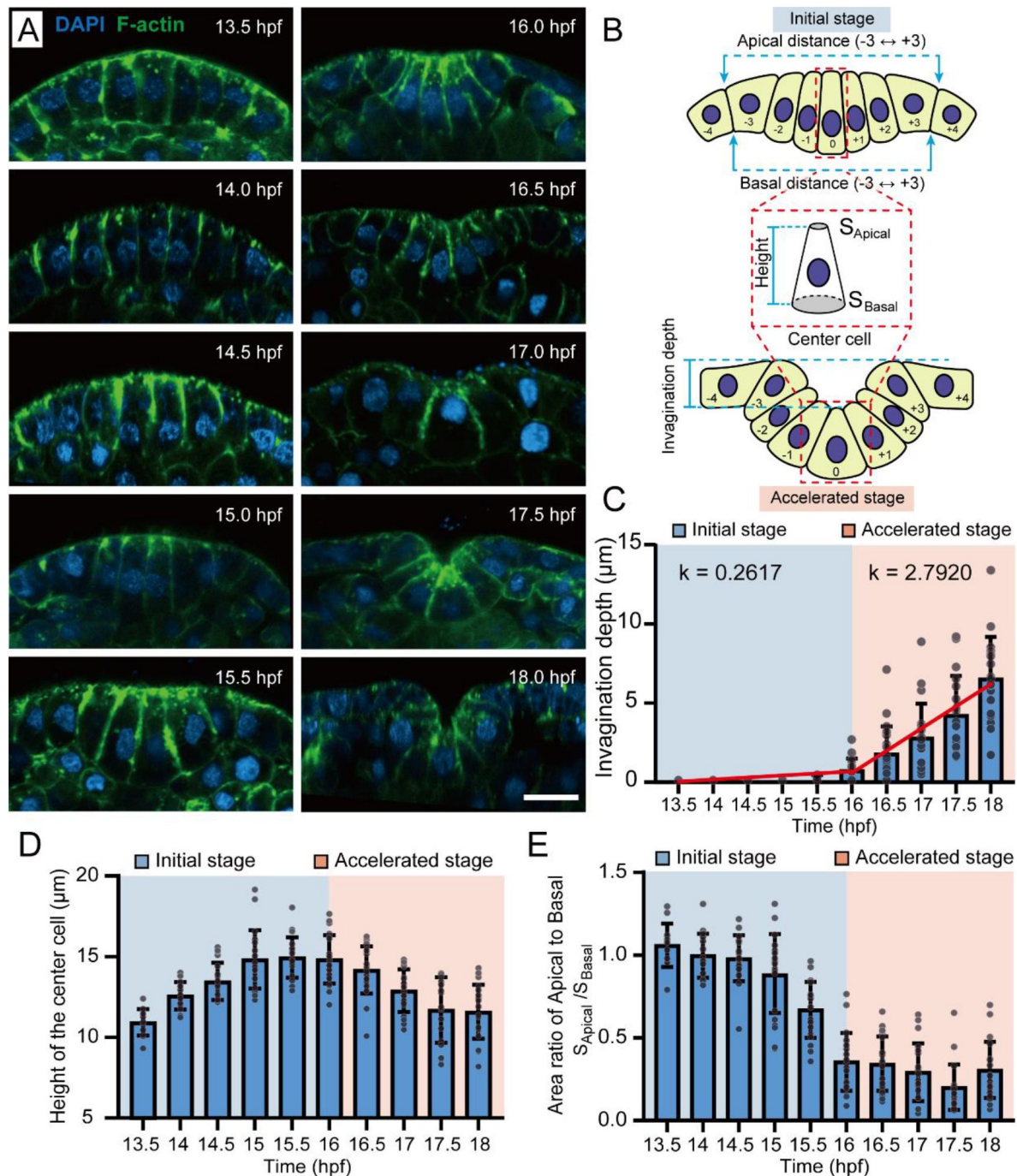


Figure 1.

Cellular processes of *Ciona* atrial siphon tube invagination

(A) Representative images of atrial siphon morphogenesis in *Ciona* embryos from 13.5 hpf to 18 hpf. Scale bar: 10 μm (B) Measurement parameters of the *Ciona* atrial siphon. The cell undergoing the most prominent apical constriction at the center of invagination was defined as the center cell (0). The adjacent cells on the left and right were defined sequentially as -1, -2, -3, -4 and +1, +2, +3, +4, respectively. (C) Quantification of the invagination depth in the atrial siphon of *Ciona* embryos. Red lines indicate linear regression fits of invagination depth during the initial (13.5-16 hpf, $k = 0.2617$) and accelerated (16-18 hpf, $k = 2.7920$) stages. $n = 20$. (D, E) Quantification of the center cell height and apical-to-basal area ratio at the atrial siphon of *Ciona* embryos. The blue-shaded region represents the initial stage, while the orange-shaded region indicates the accelerated stage. $n = 20$.

Additionally, top view imaging showed that during the initial stage, atrial siphon primordium cells were arranged in a circular pattern (**Figure 2A** [↗](#)). Active myosin exhibited a similar ring-like localization, suggesting that the early stages of atrial siphon morphogenesis are driven by circumferential contractile forces generated by phosphorylated myosin, facilitating inward constriction of the primordium (**Figure 2A, B** [↗](#)). Furthermore, quantification of the lateral cell distance (**Figure 1** [↗](#)—**figure supplement 1B** [↗](#)) demonstrated a reduction in spacing between peripheral cells, indicating that surrounding cells moved centripetally toward the center position.

Integrating the localization of actomyosin with the observed cell shape changes, we propose a hypothesis in which the bidirectional reorganization of contractile forces at the apical and lateral regions drives epithelial invagination during atrial siphon morphogenesis (**Figure 2C** [↗](#)). Initially, actomyosin translocated from the lateral regions to the apical domains, generating contractile forces for apical constriction (**Figure 1** [↗](#)—**figure supplement 1A** [↗](#)). Meanwhile, the inward compression from surrounding cells facilitates cell elongation, establishing the initial invaginated cell morphology and preparing for subsequent morphogenesis (**Figure 1** [↗](#)—**figure supplement 1B** [↗](#)). Then, actomyosin translocated from the apical domains to the lateral regions, generating contractile forces that promoted center cell shortening and accelerate the tissue invagination (**Figure 2C** [↗](#)).

Center cell height is coupled with invagination depth

To further verify our hypothesis and explore the relationship between myosin contractility, force redistribution, and the progression of invagination, we overexpressed MRLC (T18ES19E)::mCherry (a diphosphorylated mutant that enhances myosin activity) ([Espinoza-Fonseca et al., 2014](#) [↗](#)), MRLC (T18AS19A)::mCherry (an unphosphorylated mutant that reduces myosin activity) ([Iwasaki et al., 2001](#) [↗](#)) and MRLC::mCherry (wild-type control) in the atrial siphon primordium to modulate contractile force in embryos, respectively (**Figure 3A, B** [↗](#)). The results showed that the invagination initiated earlier (15 hpf) compared to control group in MRLC (T18ES19E)-expressed embryos. At 16 hpf, MRLC (T18ES19E) group exhibited a deeper invagination and shorter center cell height. In contrast, in MRLC (T18AS19A) group, the invagination initiation time was delayed. Even at 16 hpf, the center cell height did not decrease and the invagination did not occur (**Figure 3B** [↗](#)). Notably, the increase in invagination depth is strongly correlated with the reduction in center cell height, with both changes occurring synchronously across experimental groups with different myosin activities (**Figure 3B** [↗](#)). These results demonstrate a strong coupling between the reduction in center cell height and the increase in invagination depth, suggesting that this process is closely associated with the redistribution of myosin localization from the apical to lateral regions as invagination progresses.

Inhibition of myosin activity during apical-to-lateral redistribution impedes invagination progression

To further examine the necessity of the redistribution of myosin between the apical and lateral regions during invagination, we utilized an optogenetic MLCP-BcLOV4 ([Berlew et al., 2021](#) [↗](#); [Berlew et al., 2020](#) [↗](#); [Berlew et al., 2022](#) [↗](#); [Glantz et al., 2019](#) [↗](#); [Glantz et al., 2018](#) [↗](#); [Yamamoto et al., 2021](#) [↗](#)) system (**Figure 4A** [↗](#)) to specifically inhibit myosin II activity at the critical redistribution time point between the initial invagination and the accelerated invagination stage (16-17 hpf). This system has been proven to effectively reduce the contractility of epidermal cells in *Ciona* embryos ([Qiao et al., 2023](#) [↗](#)). Compared to the control group (**Figure 4** [↗](#)—**figure supplement 1** [↗](#)) and dark treatment group, the experimental group with 1 h light exposure, which activated the optogenetic system, blocked further invagination (**Figure 4B** [↗](#), **Figure 4**—video 1, 2), and while invagination ceased to deepen, the center cell height did not show a significant decrease (**Figure 4C, D** [↗](#)). These results confirm that the redistribution of myosin contractility from the apical to lateral regions is essential for the development of invagination.

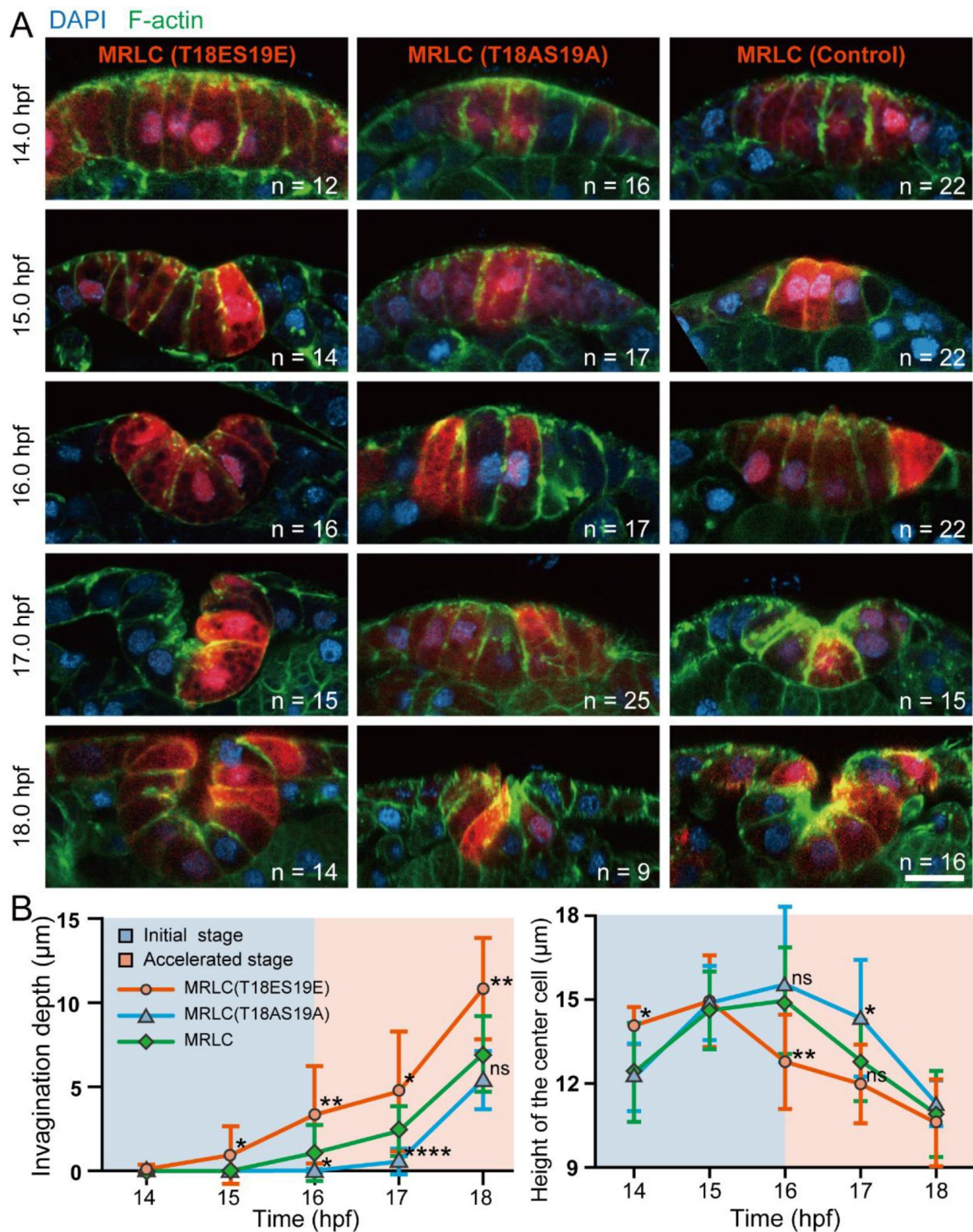


Figure 3.

Modulation of atrial siphon invagination by overexpression of myosin mutants

(A) Representative images of *Ciona* atrial siphon primordium expressing wild-type and mutant MRLC constructs. Scale bar: 10 μ m. (B) Quantification of invagination depth and the center cell height in MRLC (T18ES19E), MRLC (T18AS19A), and MRLC groups. The blue-shaded region represents the initial stage, while the orange-shaded region indicates the accelerated stage. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$.

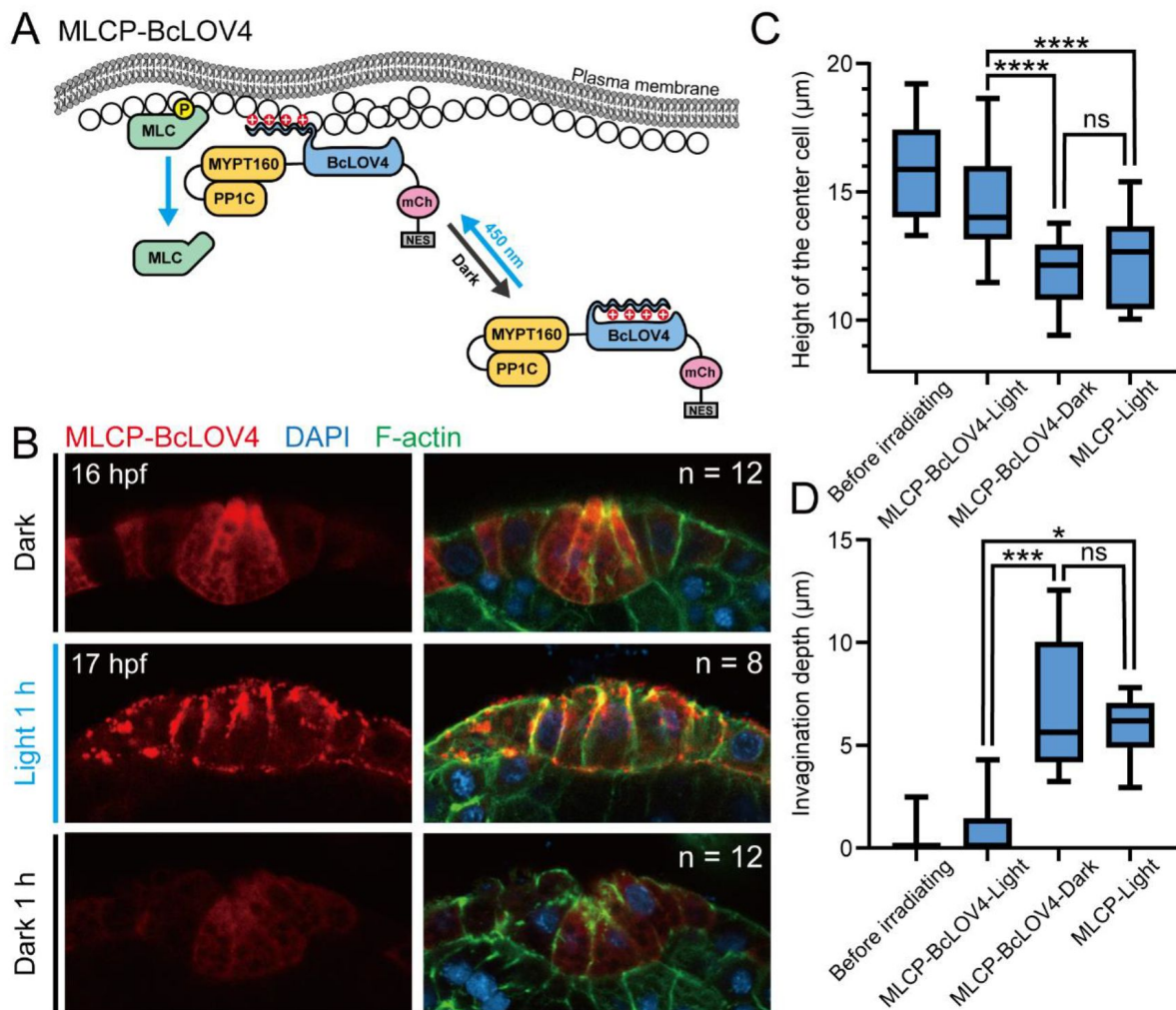


Figure 4.

Disruption of contractile forces during rapid invagination of the *Ciona* atrial siphon using an optogenetic system

(A) Schematic diagram depicting the structure and mechanism of the MLCP-BcLOV4 system. The PP1C::MYPT160::BcLOV4::mCherry::NES fusion protein is initially dispersed in the cytoplasm. Upon exposure to blue light, BcLOV4 undergoes a conformational change, allowing it to interact electrostatically with the plasma membrane. This leads to the recruitment of the PP1C and MYPT160 components, subunits of myosin light chain phosphatase (MLCP), to the membrane, where they reduce myosin activity. (B) Representative images of developmental progression in the MLCP-BcLOV4 expression group exposed to blue light for 1 h, and in the dark control group maintained in darkness for 1 h. Scale bar: 10 μm (C, D) Quantification of invagination depth and the center cell height in the MLCP-BcLOV4 expression group, the dark control group and MLCP control group (Figure 4—figure supplement 1). * $p < 0.05$, *** $p < 0.001$, **** $p < 0.0001$.

Vertex model simulations recapitulate the mechanical process of ascidian siphon tube invagination

To validate the hypothesis that the redistribution of contractile forces drives invagination, we developed a cell-based vertex model (Alt et al., 2017) and performed simulations for atrial siphon morphogenesis. The cross-section of the epithelial cells is represented by two-dimensional polygons with vertices and edges. The interaction between the epithelia and the other cells below, which function as structural support, are simplified as a basal elastic force (Figure 5A). Cell deformation and tissue morphology are simulated by the motion of vertices, which is controlled by the constraint of cell area and cortical contraction, tissue surface bending, and most importantly, active line tension dynamics (Figure 5A, see Methods). The active tension Γ_j^{abl} , on the apical, lateral and basal edges of a specific cell j , are induced by dynamic production and distribution of actomyosin m_j^{abl} on corresponding regions through a tension coefficient k_m (Figure 5B). Here we use the apical and lateral F-actin fluorescence intensities normalized by inactive basal intensity measured in experiments (Figure 1—figure supplement 1A) to describe the evolution of actomyosin intensity in the center cell of the model. The actomyosin intensity in adjacent 9 cells (-4 to $+4$) are linearized from inactive to active $m_{\text{exp}}^{\text{a,l}}$ (see Methods). We applied the periodic boundary conditions along x axis to avoid the boundary effect, and reduced the periodic length in simulations to mimic the decreased distance between adjacent cells obtained from experiments (Figure 1—figure supplement 1B). The numerical simulations were conducted based on an initial flat epithelial shape consisting of $N = 25$ cells at 13 hpf to examine the morphological processes driven by active tensions (see Methods).

Our mechanical model successfully reproduced the invagination of the atrial siphon observed in experiments (Figure 5C, D, Figure 5—video 1). During the initial stage, the rapid accumulation of actomyosin on the apical surface gave rise to apical tension. The active tension induced the contraction of central region and led the adjacent cells to migrate to the center cells, which induced the cell elongation and apicobasal imbalance, breaking the apicobasal symmetry and initializing the invagination into the basal side. After 15 hpf, the lateral actomyosin intensity increased significantly and induced high tension on the lateral surface, pulling cells and shortening the cell height. The coupling apico-lateral contraction further facilitated the invagination, which gave rise to the siphon-like epithelial shape formation at 18 hpf (Figure 5D). The invagination depth and center cell height were quantified during this process and were consistent with the experimentally measured data (Figure 5C). We further varied both apical and lateral actomyosin intensities m by introducing a scaling factor α to experimental measured m_{exp} as $m = \alpha m_{\text{exp}}$ into the model, to simulate unphosphorylated or diphosphorylated mutants whose overall myosin activities in cells are altered in experiments (Figure 3). As anticipated, simultaneous enhancement of apical and lateral actomyosin activities significantly increased invagination depth while reducing center cell height (and vice versa), closely corresponding with experimental mutant data (Figure 5E, F). These results validate the validity of our model in capturing the mechanical process of ascidian siphon tube invagination.

Coupled mechanism of apicobasal bending and lateral contraction via actomyosin redistribution

To theoretically distinguish the effects of apical and lateral contraction on morphogenesis, we performed perturbation on the intensity of regional actomyosin. The apical actomyosin intensity m^{a} was modulated as $m^{\text{a}} = \alpha_{\text{a}} m_{\text{exp}}^{\text{a}}$, where $m_{\text{exp}}^{\text{a}}$ represented the experimental data shown in Figure 5B. As a result, the invagination depth increased significantly under higher α_{a} (Figure 6A), while the central height remained relatively independent of α_{a} when $\alpha_{\text{a}} \geq 0.6$ (Figure 6B). Intriguingly, the depth value became negative when $\alpha_{\text{a}} \leq 0.2$, which meant the invagination cannot occur but was replaced by evagination, where the decrease of the central height after 15 hpf also

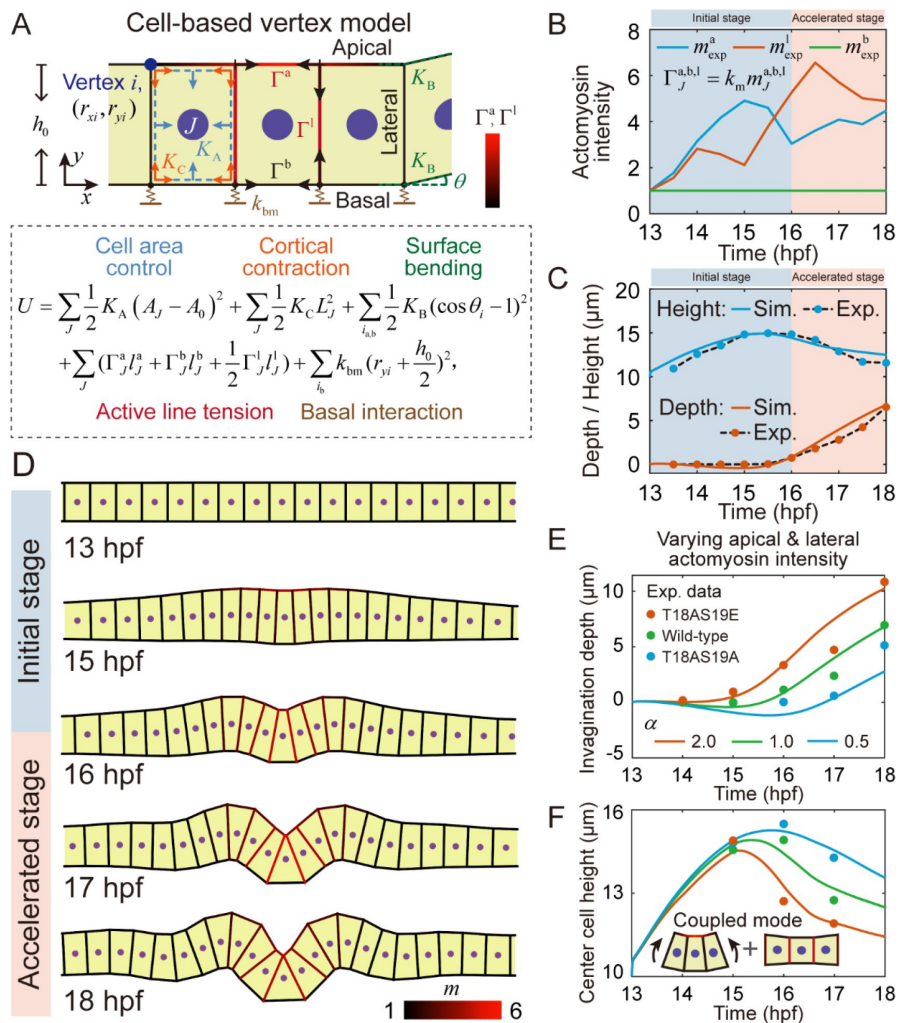


Figure 5.

Simulations and analysis based on vertex model

(A) Schematic of the cell-based vertex model. The epithelial tissue is constructed by the apical, basal, and lateral vertices and edges. The effective energy U takes into cell area constraint (modulus K_A), passive cortical contraction (coefficient K_C), tissue surface bending (modulus K_B), basal interaction (coefficient k_{bm}), and apical, basal and lateral active tensions (Γ^a, Γ^l and Γ^b , respectively). $\mathbf{r}_i = (r_{xi}, r_{yi})$ denotes the Cartesian coordinates of vertex i . A_j , L_j , and $l_j^{a,b,l}$ are the area, perimeter, and length of apical, basal, lateral edges of cell j . θ_i represents the angle between adjacent apical or basal edges connected to vertex i . A_0 is the reference area and h_0 is the initial cell height. (B) Temporal evolution of actomyosin intensities at the apical, lateral, and basal domains of the center cell in the model, parameterized based on experimentally measured F-actin distribution. The blue-shaded and orange-shaded regions represent the apical constriction stage, and the invagination stage, respectively. (C) Evolution curves of invagination depth (blue) and center cell height (orange) in simulations. Dots represent corresponding experimentally measured values. (D) Snapshots at specific timepoints during simulation progression. The colors of edges represent actomyosin intensity. (E, F) Invagination depth (E) and center cell height (F) regulated by changes in actomyosin intensities under the scaling factor α at both apical and lateral regions of all cells. The data points are from the mutant experiments in Figure 3B. Figure 5—video 1. Vertex model simulation. Edge colors represent actomyosin intensity. Bottom row (left to right):

1. Evolution curves of invagination depth;
2. Evolution curves of center cell height;
3. Temporal evolution of actomyosin intensities at the apical, lateral, and basal domains of the center cell in the model, parameterized based on experimentally measured F-actin distribution.

disappeared. This inversion of the tissue shape was attributed to the inversion of apico-basal differential of actomyosin intensity (**Figure 6C**), in which a low α_a resulted in apical actomyosin intensity falling below the reference basal intensity. These results suggest a bending mode under an equivalent moment induced by apico-basal tension imbalance, which breaks tissue symmetry and determines the buckling direction, facilitating the invagination without obviously affecting the cell height. In contrast, under modified lateral actomyosin, the central height varied distinctly (**Figure 6D**) with mild invagination depth variation (**Figure 5B**). This indicated that lateral tension primarily facilitates tissue invagination by regulating cellular contraction along the thickness direction, representing a lateral contraction mode. Collectively, these results demonstrate that the invagination of the atrial siphon is mediated by the coupled biomechanical mechanism of a bending mode induced by apicobasal tension imbalance and a contraction mode driven by lateral tension.

Finally, to clarify the characteristics of actomyosin redistribution between apical and lateral regions, we calculated the total actomyosin intensity $S_c = m_c^a + m_c^l$ and the ratio of apical and lateral intensity $R_c = m_c^a / m_c^l$ in the center cell (**Figure 6E**). During the whole morphogenesis, S_c increased persistently, providing the driving force underlying epithelial invagination. While a significant peak value in the curve of R_c was observed during the initial stage, representing the translocation from lateral to apical regions during 14-15 hpf and a reverse translocation during 15-16 hpf (**Figure 6E**). To investigate the translocation effect quantitatively, we modified the translocation level after 14 hpf in the model as $R_c = \beta (R_{\text{exp}} - 1) + 1$, where β is the translocation intensity controlling the redistribution ratio. As shown in **Figure 6F**, varying total actomyosin $S_c = \alpha S_{\text{exp}}$ and apical-lateral translocation R_c controlled by α and β , respectively, were introduced into simulations. Computational results indicated that the overall invagination magnitude (**Figure 6G**) is predominantly regulated by the total actomyosin intensity within cells, whereas stronger regional translocation and redistribution contribute to the rapid reduction in height of invaginating cells (**Figure 6H**), thereby ensuring direct cell shape changes that support robust atrial siphon formation in the ascidian.

Discussion

The morphogenesis of the *Ciona* atrial siphon provides a compelling model to dissect how biomechanical forces orchestrate tissue invagination. Our study revealed that actomyosin contractility underwent a dynamic spatial redistribution during this process, initially translocating from the lateral cortex to the apical region to promote apical constriction and establish the initial cell shape, and subsequently redistributing back to the lateral regions to promote center cell shortening and accelerate the tissue invagination. By combining actomyosin localization analysis, genetic perturbations, optogenetic manipulation, and vertex model, we established a mechanistic framework linking force dynamics to tissue remodeling. This work not only advances our understanding of siphon morphogenesis but also offers broader insights into the principles governing force-driven tissue shaping in developing organisms.

Our findings demonstrated that the atrial siphon invagination was driven by a coupled mechanism including apicobasal tension imbalance and lateral contraction in a two-stage process. During the initial stage (13.5-16.0 hpf), actomyosin translocated from the lateral cortex to the apical cortex, generated contractile forces that induced apicobasal tension imbalance and reduced apical cell area and elongated the primordium. This aligns with classical models of epithelial folding, where apical actomyosin networks drive tissue curvature through localized contraction, often accompanied by an increase in cell height, as observed in the invagination of the *Drosophila* embryonic trachea (Schottenfeld et al., 2010) and salivary gland placode (Pearl et al., 2017). Additionally, in traditional invagination models, various epithelial tissues use different mechanisms to drive deeper invagination. In *Drosophila* tracheal development, for instance, mitosis acts as a critical turning point in accelerating invagination (Kondo & Hayashi, 2013).

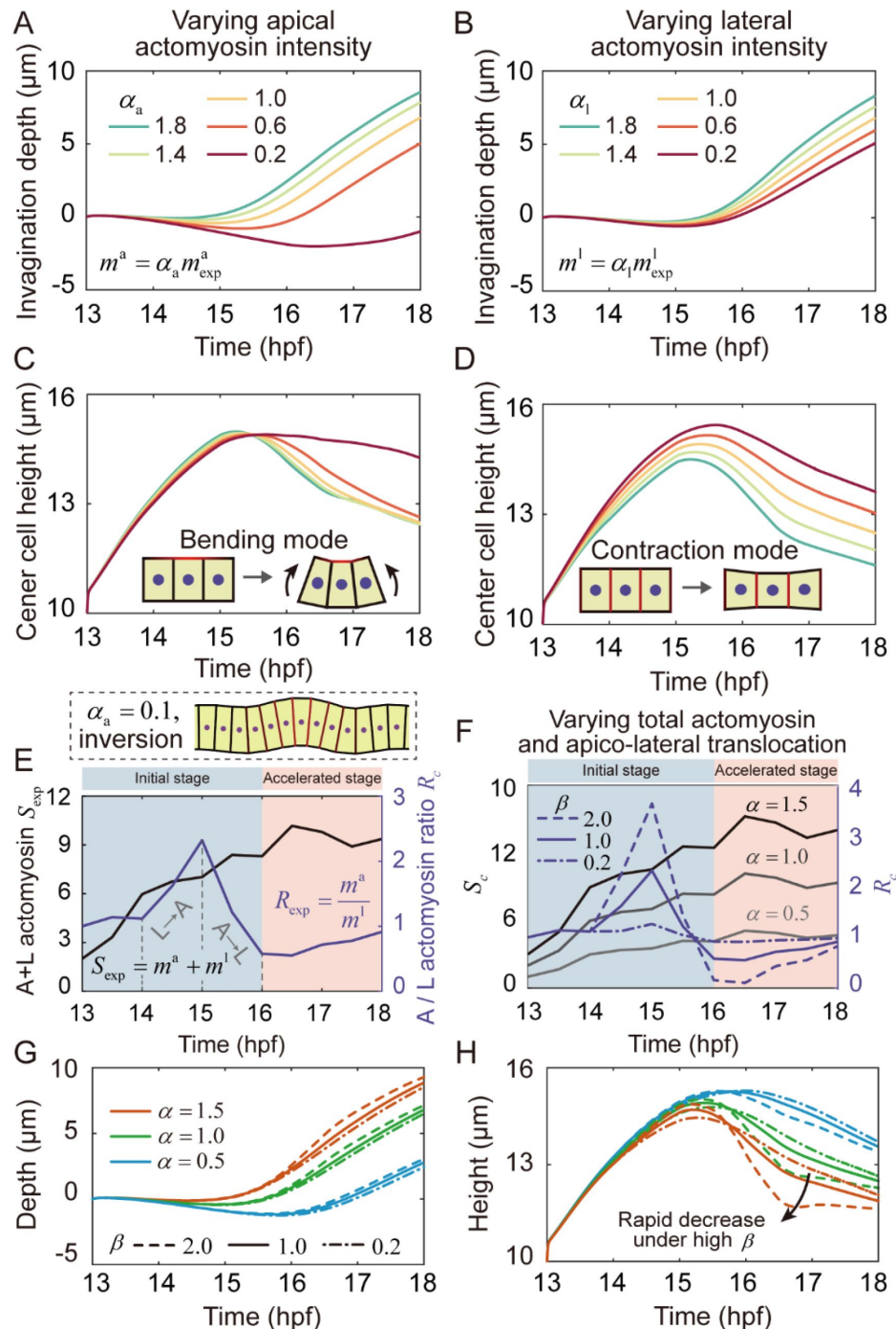


Figure 6.

Mechanics of epithelial invagination revealed by vertex model simulations

(A, C) Invagination depth (A) and center cell height (C) under varying apical actomyosin intensity represented by a scaling factor α_a , with the lateral intensity unchanged. A Schematic illustration of the bending mode induced by apicobasal imbalance is presented. A snapshot of a specific invagination modeling under $\alpha_a = 0.1$ is shown. (B, D) Invagination depth (B) and center cell height (D) under varying lateral actomyosin intensity represented by a scaling factor α_l , with the apical intensity unchanged. A Schematic illustration of the contraction mode induced by lateral tension is presented. (E) Temporal evolution of the sum S_{exp} and ratio R_{exp} of apical and lateral actomyosin intensities in the center cell in experiments. (F) Varying sum S_c and ratio R_c of actomyosin intensities regulated by α and β in simulations. (G, H) Invagination depth (G) and center cell height (H) under varying α and β .

During the early slow phase of tracheal invagination, apical constriction under EGFR signaling, forms a shallow pit. As invagination transitions into a rapid phase, mitotic entry induces cell rounding, which increases tension and epithelial buckling, thereby accelerating the invagination process and facilitating the internalization of placode cells (Kondo & Hayashi, 2013 [↗](#); Nishimura et al., 2007 [↗](#)). Moreover, cell apoptosis can also expedite the invagination process, as seen in *Drosophila* leg morphogenesis, where actomyosin cables formed in apoptotic cells help trigger the invagination (Kiehart, 2015 [↗](#); Manjón et al., 2007 [↗](#); Monier et al., 2015 [↗](#)). Distinct from these classical models, the accelerated invagination stage of *Ciona* atrial siphon morphogenesis was driven by a critical redistribution of apical actomyosin to the lateral regions, during accelerated stage (16.0–18.0 hpf), providing a force-generating mechanism for invagination progression without relying on cell proliferation or apoptosis to accelerate the process. The lateral enrichment of p-MLC correlated with a reduction in center cell height and the initiation of invagination, suggesting that lateral forces actively pulled the tissue inward. This spatial redistribution of contractile forces in *Ciona* atrial siphon invagination is reminiscent of mechanisms observed in certain epithelial folding processes, such as the *Drosophila* wing disc, where invagination is not solely driven by apical constriction but can also result from increased lateral tension or reduced basal tension (Sui et al., 2018 [↗](#)). Our optogenetic inhibition of myosin activity during the apical-to-lateral redistribution—using the MLCP-BcLOV4 system—directly confirmed that this redistribution was not merely a passive consequence of invagination but was an active driver of the process.

Our study revealed a strong coupling between cellular shape changes and tissue remodeling during *Ciona* atrial siphon invagination, where the reduction in center cell height was closely associated with invagination depth. Tissue morphogenesis emerges from the integrated actions of multiple cellular behaviors, such as contraction, adhesion, and migration, that work in concert to remodel epithelial architecture (Chan et al., 2017 [↗](#)). The transmission of local cell-generated forces across tissues is further shaped by mechanical cues and geometric constraints, enabling the transformation of cellular-scale changes into organ-scale morphogenesis (Collinet & Lecuit, 2021 [↗](#)). By perturbing myosin activity, we found that this coupling was directly regulated: dominant-negative myosin mutants delayed both processes, whereas hyperactive myosin accelerated invagination. The quantitative agreement between vertex model analysis and experimental measurements validated the critical role of actomyosin contractility dynamics in linking cellular shape changes to tissue remodeling. Furthermore, the model also uncoupled the effects of the actomyosin in different domains, and predicted that stronger apical-lateral translocation facilitates the rapid reduction in height of invaginating cell. The *Ciona* atrial siphon, which underwent invagination without cell proliferation or apoptosis, provided a minimal system for investigating the role of biomechanical forces in tissue remodeling. Our findings emphasized that the deformation of a few key cells under spatially precise mechanical regulation, along with their coordination with surrounding tissues, is crucial for proper morphogenesis.

While this study clarified the role of actomyosin redistribution in atrial siphon invagination, several open questions remain regarding the molecular mechanisms underlying this process. One key question is what molecular signals drive the apical-to-lateral redistribution of contractility. Possible candidates include Rho GTPase pathways (Martin et al., 2016 [↗](#)), which regulate myosin activation; apical-basal polarity (Peng et al., 2020 [↗](#)), which coordinate actin reorganization during morphogenesis; and mechanosensitive ion channels that respond to tissue strain (Jin et al., 2020 [↗](#)). Additionally, the role of cell-cell adhesion and extracellular matrix remodeling in mediating force transmission during invagination requires further investigation. Using tension sensors or perturbing adhesion molecules may provide new insights into these interactions. Although our vertex model effectively predicts the contribution of force redistribution to invagination, the spatiotemporal evolution of actomyosin could also be affected by mechanosensitive pathways while generating active contraction in morphogenesis, which constructs a bidirectional loop of actomyosin regulation. Future theoretical developments should focus on integrating mechanochemical feedback linking molecular signaling with tissue-level deformation to explain robust morphogenesis.

Materials and Methods

Experimental animals' preparation and electroporation

Ciona adults were collected from the coast of Qingdao and Rongcheng, Shandong, China. They were maintained in the laboratory seawater circulation system under constant light conditions to maintain their stability. Mature eggs and sperm were separately collected from the dissected adults and subsequently mixed for fertilization. The Fertilized eggs were dechorionated in seawater containing 1 % sodium thioglycolate (T0632; Sigma), 0.05 % protease E (P5147; Sigma) and 0.032 M NaOH. The dechorionated eggs were then used for plasmid electroporation based on the previous technique procedure(Christiaen et al., 2009 [DOI](#)). Finally, the embryos were cultured in an agar-coated dish with microporous-filtered seawater in 18°C for further observation. Detailed key resource information is annotated in Appendix 1—table 1.

Plasmid construction

Ciona myosin II light chain (MLRC) was amplified with primers MRLC-F and MRLC-R (Table S1). MRLC-mCherry(Dong et al., 2011 [DOI](#)); MRLC(T18AS19A)-mCherry and MRLC(T18ES19E)-mCherry(Denker et al., 2015 [DOI](#)); MLCP-BcLOV4 and MLCP(Qiao et al., 2023 [DOI](#)) were amplified with the primers listed in Appendix 1—table 2.

Immunofluorescence

Ciona embryos were fixed with stationary liquid(Sherrard et al., 2010 [DOI](#)), which consisted of 100 mM HEPES (pH 6.9); 100 mM EGTA (pH 7.0); 10 mM MgSO₄; 2% formaldehyde; 0.1% glutaraldehyde; 300 mM dextrose and 0.2% Triton X-100 for 40 m at room temperature. Following fixation, the embryos were washed three times with PBS and subsequently incubated in PBST (PBS supplemented with 0.1% Triton X-100) for 30 m to enhance permeability. To reduce autofluorescence, embryos were treated with 0.1% sodium borohydride in PBS for 20 m at room temperature. For immunostaining, embryos were incubated with a 1:250 dilution of Phospho-Myosin Light Chain 2 (Ser19) antibody (#3671; Cell Signaling) at room temperature for 24 h. After three additional washes with PBS, a 1:200 dilution of Alexa Fluor 568 anti-Rabbit IgG (A11011; Invitrogen) was added and incubated at room temperature for 48 h. For cell boundary visualization, embryos were stained with Alexa Fluor 488 Phalloidin (A12379; Invitrogen) at a 1:200 dilution. Finally, after three washes with PBS, embryos were mounted in DAPI-containing mounting medium and prepared for imaging.

Imaging and optogenetics

Live imaging, photoactivation experiments, and image acquisition were performed using a Zeiss LSM 980 confocal microscope (Carl Zeiss). For optogenetic experiments, *Ciona* embryos were placed in a 35 mm glass-bottom dish for imaging. To activate the optogenetic system, the designated region of interest was exposed to a 488 nm laser for 1 h. The control group for dark treatment was placed in a dark box at the same temperature for 1 h. All images were analyzed and quantified using ImageJ (version 1.54p, NIH) and Imaris (version 9.0.1, Bitplane).

Vertex model simulations

Model description

The epithelial cells are represented by two-dimensional polygons with vertices and edges. Vertices on the apical surface of cells are subject to no boundary constraints, while vertices on the basal surface interact with basal spring forces to simulate the structural support provided by subjacent

cells (**Figure 5A** [↗](#)). The potential energy of the tissue is formulated by

$$U = \sum_J \frac{1}{2} K_A (A_J - A_0)^2 + \sum_J \frac{1}{2} K_C L_J^2 + \sum_{i,s} \frac{1}{2} K_B (\cos \theta_i - 1)^2 + \sum_J (\Gamma_J^a l_J^a + \Gamma_J^b l_J^b + \frac{1}{2} \Gamma_J^l l_J^l) + \sum_{i,s} k_{bm} (r_{xi} + \frac{h_0}{2})^2, \quad (1)$$

where the six terms result from, respectively, cell area control, cell cortex contractility, surface bending energy, active line tensions on apical, basal, and lateral surfaces, and subjacent interactions. K_A is the cell area modulus, and A_J and A_0 are the current and preferred areas of the J th cell, respectively. K_C denotes the passive cortical contraction coefficient of cells and L_J represents the cell perimeter. K_B denotes the bending modulus of the monolayer and θ_i is the angle between the neighboring apical or basal edges of vertex i . Γ_J^a , Γ_J^b , and Γ_J^l are the active line tensions acting on the apical, basal, and lateral surfaces of the J th cell with edge lengths l_J^a , l_J^b and l_J^l , respectively. k_{bm} is the elastic coefficient of basal interaction and h_0 is the initial cell height. $\mathbf{r}_i = (r_{xi}, r_{yi})$ denotes the Cartesian coordinates of vertex i . $\sum_J$ stands for the summation over all cells and $\sum_{i,s}$ runs over all apical and basal vertices.

To describe the cell actomyosin evolution during the morphogenesis, the line tensions Γ_J^a , Γ_J^b , and Γ_J^l are proportional to actomyosin levels on the apical, basal, and lateral surfaces of the cell J as

$$\Gamma_J^a = k_m m_J^a, \Gamma_J^b = k_m m_J^b, \Gamma_J^l = k_m m_J^l, \quad (2)$$

where k_m is the tension coefficient based on the myosin intensity m_J^a , m_J^b and m_J^l . Given that the intensity of F-actin in the center cell has been measured as $m_{exp}^{ab,l}$ during the development, we define $m_{exp}^{ab,l}$ as the actomyosin intensity of the center cell in the model (**Figure 5B** [↗](#)).

Simulation scheme

In the simulations, we set the cell number $N = 25$ and a periodic boundary condition in the x direction with periodic box length $L = NA_0 / h_0$. The central nine cells (index -4 to $+4$) are considered as active cells under contractile forces whose actomyosin intensity varies in different stages. The other cells are observed to be inactive whose actomyosin intensities are always equal to $m_0 = m_{exp}^b = 1$. The actomyosin intensities of the central nine cells (-4 to $+4$) are calculated through linear interpolation between inactive m_0 of boundary cells and active $m_{exp}^{ab,l}$ of the most center cell according to the distance. The periodic length L in simulations is reduced slowly to mimic the decreased distance between adjacent cells obtained from experiments (**Figure 1** [↗](#)—**figure supplement 1B** [↗](#)). In our simulations, tissue morphogenesis is represented by vertices motion and cell deformation. The motion of vertex i obeys the Langevin equation

$$\eta \frac{d\mathbf{r}_i}{dt} = -\frac{\partial U}{\partial \mathbf{r}_i}, \quad (3)$$

where η denotes the damping coefficient. The equation was numerically solved using the forward Euler method with a time step of 0.01, using MATLAB R2021a. Simulations are run until $t = 400$, which corresponds to 18 hpf in experimental development with the time scale $\tau = 45$ s. The length scale $\varepsilon = 8.91 \mu\text{m}$ is determined by the normalization of initial height h_0 in model under real center cell height. The other parameters are all normalized with the time scale $\tau = 45$ s, length scale $\varepsilon = 8.91 \mu\text{m}$, and damping coefficient η . Dimensionless parameters are set as follows: $K_A = 1$, $K_C = 0.01$, $K_B = 0.03$, $k_{bm} = 0.1$, $k_m = 0.002$, $A_0 = 1$, and $h_0 = 1.25$. These parameter values are determined by numerical simulations under which the simulated invagination depth and center cell height imitates the experimental observations.

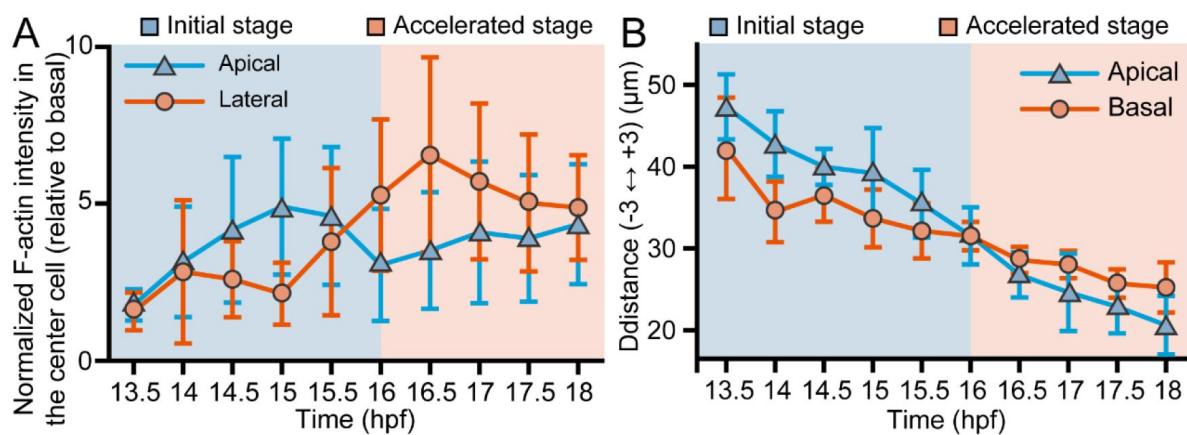


Figure 1 Figure supplement 1

Quantification of F-actin intensity and intercellular distance during *Ciona* atrial siphon morphogenesis

(A) Normalized F-actin intensity at the apical and lateral regions of center cells during *Ciona* atrial siphon morphogenesis (basal level set to 1). (B) Quantification of the linear distance between the -3/-4 and +3/+4 cell junctions at the apical or basal surface in the atrial siphon of *Ciona* embryos. The blue-shaded region represents the initial stage, while the orange-shaded region indicates the accelerated stage. Representative images are shown in [Figure 1A](#). n = 20.

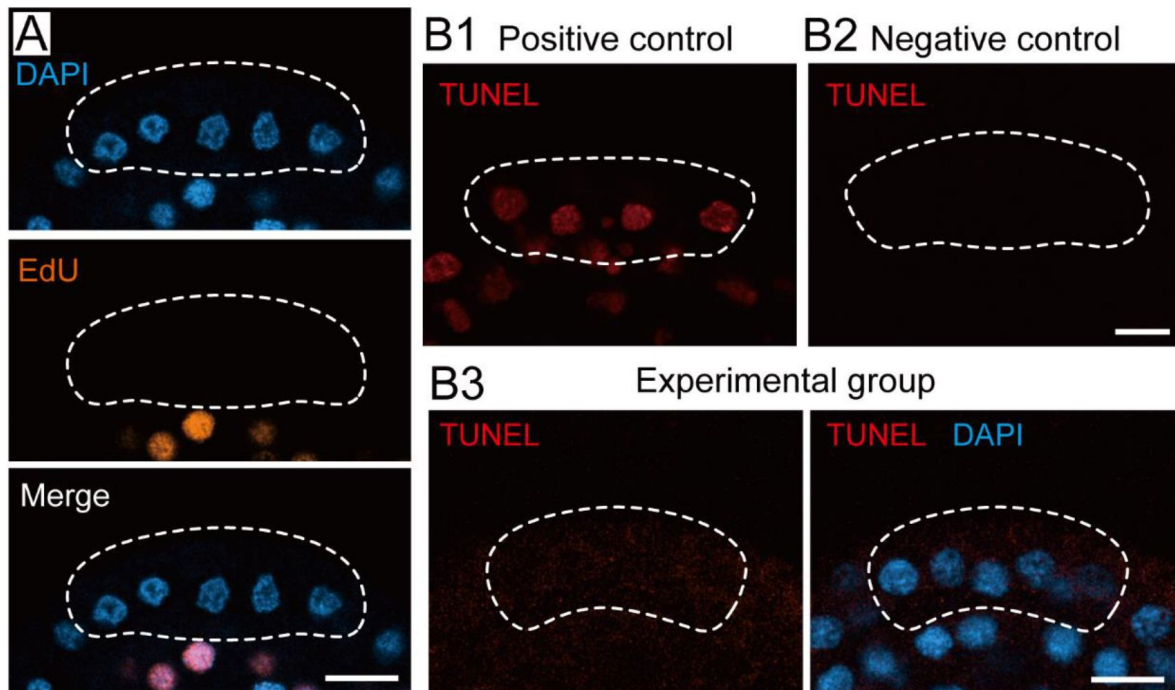


Figure 1 figure supplement 2

EdU and TUNEL staining during *Ciona* atrial siphon morphogenesis

(A) Representative images of EdU staining at 14-15 hpf. Orange: EdU-positive nuclei indicating cell proliferation. Blue: DAPI. No EdU signal was detected in the atrial siphon primordium (white dashed outline). Scale bar: 10 μ m. n = 10. (B1-3) Representative images of TUNEL staining at 15 hpf. (B1) Positive control: DNase I pretreatment (20 U/mL, 10 min) induced DNA fragmentation. (B2) Negative control: staining performed without terminal deoxynucleotidyl transferase (TdT) enzyme. (B3) Experimental group: no detectable TUNEL signal in the atrial siphon primordium (white dashed outline). Scale bar: 10 μ m. n = 10.

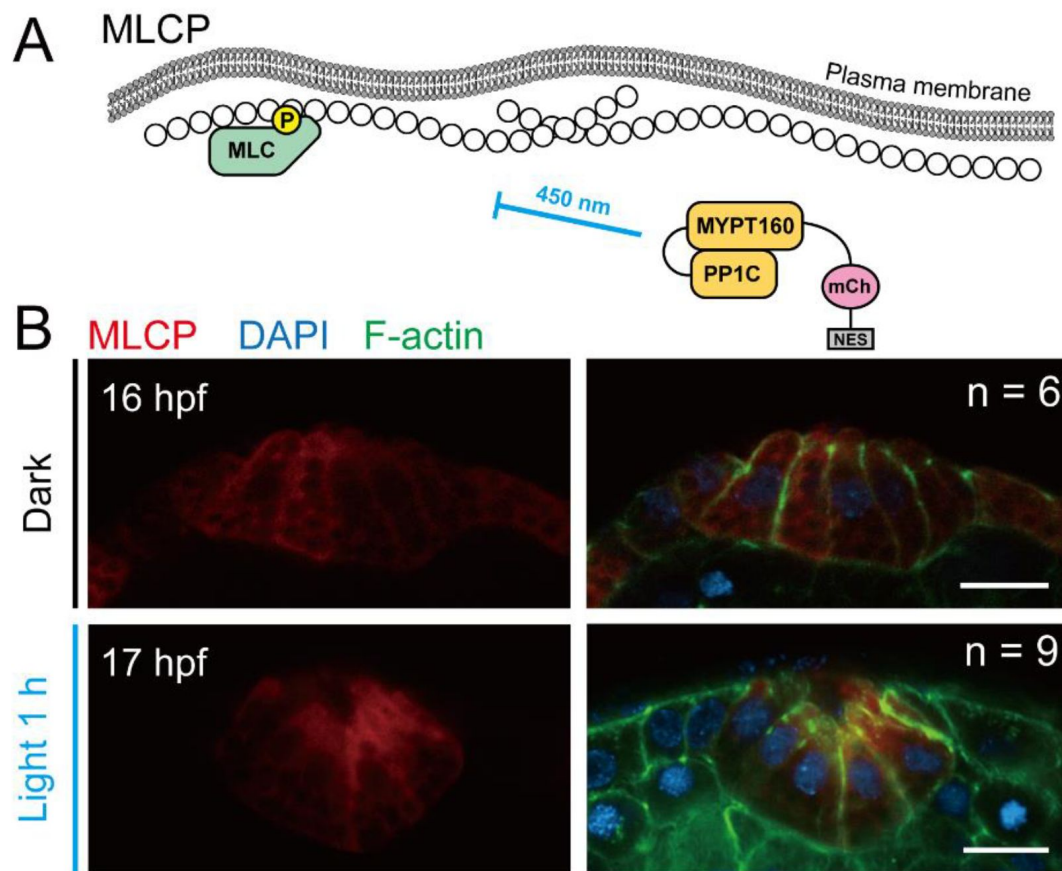


Figure 4 supplement 1

MRCL control group in the optogenetic experiment

(A) Schematic diagram depicting the structure and mechanism of the MLCP control system. The PP1C::MYPT169::mCherry::NES fusion protein remained diffuse in the cytoplasm under light exposure and failed to function.

(B) Representative images of developmental progression in the MLCP control group exposed to blue light for 1 h. Scale bar: 10 μ m. Figure 4—video 1. The developmental processes of the MLCP control group under blue light illumination for 1 h. Scale bar: 10 μ m. Figure 4—video 2. The developmental processes of MLCP-BcLOV4-expressed group under blue light illumination for 1 h. Scale bar: 10 μ m.

REAGENT or RESOURCE	SOURCE	IDENTIFIER
antibodies		
Phospho-Myosin Light Chain 2 (Ser19) Antibody	Cell Signaling	#3671
Alexa Fluor 647 anti-Mouse IgG	Invitrogen	#A28181
Chemicals		
EdU Cell Proliferation Kit with AF555	BeyoClick	C0075S
Alexa Fluor™ 488 Phalloidin	Thermo fisher	A12379
Protease E	sigma-Aldrich	P5147
Sodium thioglycolate	sigma-Aldrich	T0632
D-mannitol	sigma-Aldrich	M4125
MS-222	sigma-Aldrich	E10521
VECTASHIELD® with DAPI	Vectorlabs	H1200
Phanta Max Super-Fidelity DNA Polymerase	Vazyme	P505
ClonExpress MultiS One Step Cloning Kit	Vazyme	C113
TUNEL BrightRed Apoptosis Detection Kit	Vazyme	A113
Plasmids		
Epi1> MRLC::mCherry	This study	N/A
Epi1> MRLC(T18AS19A)::mCherry		N/A
Epi1> MRLC(T18ES19E)::mCherry		N/A
Epi1>PP1c::MYPT169::BcLOV4::mCherry::NES		N/A
Epi1>PP1c::MYPT169::mCherry::NES		N/A
Software		
ImageJ	NIH Image	https://imagej.nih.gov/ij/
Prism 10 CN	Graphpad Software	https://www.graphpad.com/
Imaris 10	Oxford Instruments	https://imaris.oxinst.com/
MATLAB R2021a	MathWorks	https://www.mathworks.com/

Appendix 1—table 1

Key resource table

Construct name	Primers	Sequence (5'→3')
MRLC	Forward	ATGTCG-AGCCGACGAACTAAA
	Reverse	CTAAATGTCATCTTTTCTTTAGCG
MRLC(T18AS19A)	Forward	TGCAGCCGCCAATGTCTTTGCTATGTTGATCAA GC
	Reverse	GACATTGGCGGCTGCACGCTGGGCGCGC
MRLC(T18ES19E)	Forward	TGCAGAAGAAAATGTCTTTGCTATGTTGATCAA GCT
	Reverse	AGACATTTTCTTCTGCACGCTGGGCGCGC
BcLOV4	Forward	ATGGCCACAGACGCAATCG
	Reverse	GATTATGATCTAGAGTC
PP1c	Forward	ATGGCGGACGGGGAGC
	Reverse	CCTTTTCTTCGGCGGA
MYPT169	Forward	ATGAAGATGGCGGACGCG
	Reverse	AGCTGCTTCTATATCAACCCCTTG

Appendix 1—table 2

Primer sequences used in this study

Data and resource availability

Lead contact

Requests for further resources should be directed to, and will be fulfilled by, the lead contact, Bo Dong (bodong@ouc.edu.cn).

Materials availability

All unique/stable reagents generated in this study are available from the lead contact with a completed materials transfer agreement.

Data and code availability

Model code and data are available upon request, and any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

Acknowledgements

We are grateful to all members of the Fang Zongxi center for helpful discussions.

Additional information

Author contributions

B.D., B.L., J.Q., P.Y. conceived of the study; B.D., J.Q., P.Y., B.L., H.P., and W.S. analyzed data; P.Y. and B.L. designed the vertex model; B.D., J.Q., P.Y. and B.L. wrote the original draft. All authors approved the final version of the article.

Funding

This work was supported by the Science & Technology Innovation Project of Laoshan Laboratory (Nos. LSKJ202203204), the National Key Research and Development Program of China (2022YFC2601302), and the Taishan Scholar Program of Shandong Province, China (B.D.).

Additional files

Figure 4—video 1 [↗](#)

Figure 4—video 2 [↗](#)

Figure 5—video 1 [↗](#)

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Reviewer #1 (Public review):

Summary:

This paper investigates the physical basis of epithelial invagination in the morphogenesis of the ascidian siphon tube. The authors observe changes in actin and myosin distribution during siphon tube morphogenesis using fixed specimens and immunohistochemistry. They discover that there is a biphasic change in the actomyosin localization that correlates with changes in cell shapes. Initially, there is the well-known relocation of actomyosin from the lateral sides to the apical surface of cells that will invaginate, accompanied by a concomitant

lengthening of the central cells within the invagination, but not a lot of invagination. Coincident with a second, more rapid, phase of invagination, the authors see a relocalization of actomyosin back to the lateral sides of the cells. This 2nd "bidirectional" relocation of actin appears to be important because optogenetic inhibition of myosin in the lateral domain after the initial invaginations phase resulted in a block of further invagination. Although not noted in the paper, that the second phase of siphon invagination is dependent on actomyosin is interesting and important because it has been shown that during *Drosophila* mesoderm invagination that a second "folding" phase of invagination is independent of actomyosin contraction (Guo et al. *elife* 2022), so there appear to be important differences between the *Drosophila* mesoderm system and the ascidian siphon tube systems.

Using the experimental data, the authors create a vertex model of the invagination, and simulations reveal a coupled mechanism of apicobasal tension imbalance and lateral contraction that creates the invagination. The resultant model appears to recapitulate many aspects of the observed cell behaviors, although there are some caveats to consider (described below).

Strengths:

The studies and presented results are well done and provide important insights into the physical forces of epithelial invagination, which is important because invaginations are how a large fraction of organs in multicellular organisms are formed.

Weaknesses:

- (1) This reviewer has concerns about two aspects of the computational model. First, the model in Figure 5D shows a simulation of a flat epithelial sheet creating an invagination. However, the actual invagination is occurring in a small embryo that has significant curvature, such that nine or so cells occupy a 90-degree arc of the 360-degree circle that defines the embryo's cross-section (e.g., see Figure 1A). This curvature could have important effects on cell behavior.
- (2) The second concern about the model is that Figure 5 D shows the vertex model developing significant "puckering" (bulging) surrounding the invagination. Such "puckering" is not seen in the *in vivo* invagination (Figure 1A, 2A). This issue is not discussed in the text, so it is unclear how big an issue this is for the developed model, but the model does not recapitulate all aspects of the siphon invagination system.
- (3) In Figure 2A, Top View, and the schematic in Figure 2C, the developing invagination is surrounded by a ring of aligned cell edges characteristic of a "purse string" type actomyosin cable that would create pressure on the invaginating cells, which has been documented in multiple systems. Notably, the schematic in Figure 2C shows myosin II localizing to aligned "purse string" edges, suggesting the purse string is actively compressing the more central cells. If the purse string consistently appears during siphon invagination, a complete understanding of siphon invagination will require understanding the contributions of the purse string to the invagination process.
- (4) The introduction and discussion put the work in the context of work on physical forces in invagination, but there is not much discussion of how the modeling fits into the literature.

<https://doi.org/10.7554/eLife.108588.1.sa3>

Reviewer #2 (Public review):

Summary:

The authors propose that bidirectional translocation of actomyosin drives tissue invagination in *Ciona* siphon tube formation. They suggest a two-stage model where actomyosin first accumulates apically to drive a slow initial invagination, followed by translocation to lateral domains to accelerate the invagination process through cell shortening. They have shown that actomyosin activity is important for invagination - modulation of myosin activity through expression of myosin mutants altered the timing and speed of invagination; furthermore, optogenetic inhibition of myosin during the transition of the slow and fast stages disrupted invagination. The authors further developed a vertex model to validate the relationship between contractile force distribution and epithelial invagination.

Strengths:

- (1) The authors employed various techniques to address the research question, including optogenetics, the use of MRLC mutants, and vertex modelling.
- (2) The authors provide quantitative analyses for a substantial portion of their imaging data, including cell and tissue geometry parameters as well as actin and myosin distributions. The sample sizes used in these analyses appear appropriate.
- (3) The authors combined experimental measurements with computer modeling to test the proposed mechanical models, which represents a strength of the study. It provides a framework to explore the mechanical principles underlying the observed morphogenesis.

Weaknesses:

- (1) The concept of coordinated and sequential action of apical and lateral actomyosin in support of epithelial folding has been documented through a combination of experimental and modeling approaches in other contexts, such as ascidian endoderm invagination (PMID: 20691592) and gastrulation in *Drosophila* (PMIDs: 21127270, 22511944, 31273212). While the manuscript addresses an important question, related findings have been reported in these previous studies. This overlap reduces the degree of novelty, and it remains to be clarified how their work advances beyond these prior contributions.
- (2) One of the central statements made by the authors is that the translocation of actomyosin between the apical and lateral domains mediates invagination. The use of the term "translocation" infers that the same actomyosin structures physically move from one location to another location, which is not demonstrated by the data. Given the time scale of the process (several hours), it is also possible that the observed spatiotemporal patterns of actomyosin intensity result from sequential activation/assembly and inactivation/disassembly at specific locations on the cell cortex, rather than from the physical translocation of actomyosin structures over time.
- (3) Some aspects of the data on actomyosin localization require further clarification. (1) The authors state that actomyosin translocation is bidirectional, first moving from the lateral domain to the apical domain; however, the reduction of the lateral actomyosin at this step was not rigorously tested. (2) During the slow invagination stage, it is unclear whether myosin consistently localizes to the apical cell-cell borders or instead relocalizes to the medioapical domain, as suggested by the schematic illustration presented in Figure 2C. (3) It is unclear how many cells along the axis orthogonal to the furrow accumulate apical and lateral myosin.
- (4) The overexpression of MRLC mutants appears to be rather patchy in some cases (e.g., in Figure 3A, 17.0 hpf, only cells located at the right side of the furrow appeared to express MRLC T18ES19E). It is unclear how such patchy expression would impact the phenotype.

(5) In the optogenetic experiment, it appears that after one hour of light stimulation, the apical side of the tissue underwent relaxation (comparing 17 hpf and 16 hpf in Figure 4B). It is therefore unclear whether the observed defect in invagination is due to apical relaxation or lack of lateral contractility, or both. Therefore, the phenotype is not sufficient to support the authors' statement that "redistribution of myosin contractility from the apical to lateral regions is essential for the development of invagination".

(6) The vertex model is designed to explore how apical and lateral tensions contribute to distinct morphological outcomes. While the authors raise several interesting predictions, these are not further tested, making it unclear to what extent the model provides new insights that can be validated experimentally. In addition, modeling the epithelium as a flat sheet and not accounting for cell curvature is a simplification that may limit the model's accuracy. Finally, the model does not fully recapitulate the deeply invaginated furrow configuration as observed in a real embryo (comparing 18 hpf in Figure 5D and 18 hpf in Figure 1A) and does not fully capture certain mutant phenotypes (comparing 18 hpf in Figure 5F and 18 hpf in Figure 3B right panel).

<https://doi.org/10.7554/eLife.108588.1.sa2>

Reviewer #3 (Public review):

Summary:

In this manuscript by Qiao et al., the authors seek to uncover force and contractility dynamics that drive tissue morphogenesis, using the *Ciona* atrial siphon primordium as a model. Specifically, the authors perform a detailed examination of epithelial folding dynamics. Generally, the authors' claims were supported by their data, and the conceptual advances may have broader implications for other epithelial morphogenesis processes in other systems.

Strengths:

The strengths of this manuscript include the variety of experimental and theoretical methods, including generally rigorous imaging and quantitative analyses of actomyosin dynamics during this epithelial folding process, and the derivation of a mathematical model based on their empirical data, which they perturb in order to gain novel insights into the process of epithelial morphogenesis.

Weaknesses:

There are concerns related to wording and interpretations of results, as well as some missing descriptions and details regarding experimental methods.

<https://doi.org/10.7554/eLife.108588.1.sa1>

Author response:

Reviewing Editor Comments:

Based on the feedback from the reviewers, a focus on the following major points has the potential to improve the overall assessment of the significance of the findings and the strength of the evidence:

(1) It would be helpful to clearly articulate how these findings advance the field beyond what has already been demonstrated or suggested in other systems.

We will revise the Introduction and Discussion to better contextualize our findings. We will provide a careful comparison of the Ciona atrial siphon invagination with the other established systems to elucidate the unique aspects of our model. Highlighting our discovery of a novel bidirectional "lateral-apical-lateral" contractility as a distinct mechanical paradigm for sequential morphogenesis.

(2) It would be helpful to clarify the meaning of "translocation" and more explicitly describe the temporal and spatial patterns of active myosin localization during the two steps of invagination.

We will replace "translocation" with the more accurate and conservative term "redistribution" throughout the manuscript, including in the title. We will also revise the text in Result and Discussion sections to avoid overinterpretation. To provide a more explicit description of the spatiotemporal patterns, we will add new quantitative analyses of active myosin intensity from earlier time points (13-14 hpf) to rigorously support the initial lateral-to-apical redistribution phase. Then, we will add high-resolution top-view images to unambiguously show the ring-like localization of myosin at the apical cell-cell junctions during the initial stage. Finally, we will correct the schematic in Figure 2C to accurately reflect the predominant localization of active myosin at the apical cell-cell borders.

(3) It would be helpful to explain how the optogenetic data support the conclusion that "redistribution of myosin contractility from the apical to lateral regions is essential for the development of invagination".

We acknowledge the limitation of the original global inhibition experiment. We will perform additional experiments that combine optogenetic inhibition with subsequent immunostaining of the active myosin. By quantitatively comparing the distribution of actomyosin in light-stimulated versus dark-control embryos, we will be able to demonstrate whether the inhibition prevents the establishment of the lateral contractility domain. This will allow us to refine our conclusion.

(4) It would be helpful to describe how the modeling work fits within the existing literature on modeling epithelial folding and to address discrepancies between the model and the actual biological observations, such as tissue curvature, limited invagination depth in the model, and the "puckering" surrounding the invagination. In addition, certain descriptions of the modeling results should be clarified, as suggested by Reviewer #3.

We fully agree that we should discuss the existing theoretical work on epithelial folding more clearly. Clarifying how physical forces contribute to invagination is central to interpret the underlying mechanisms, and we appreciate the opportunity to better connect our framework to existing studies. In the revision, we will expand the Introduction and Discussion to place our model in the appropriate theoretical context and highlight how it relates to and differs from previous approaches. At the same time, we will extend the model to a curved geometric framework to more accurately reproduce the experimental observations, which will improve its predictive value. We will also revise the descriptions and schematic representations of the modeling results to enhance clarity and better align them with the biological data.

(5) It would be helpful to elaborate on the methods for quantitative image analysis and statistical tests.

We will thoroughly expand the Methods section to provide a detailed step-by-step description of image quantification procedures, including precise definitions of the apical, lateral, and basal domains used for intensity measurements and the measurement of cell surface areas and invagination depths.

Reviewer #1 (Public review):

Summary:

*This paper investigates the physical basis of epithelial invagination in the morphogenesis of the ascidian siphon tube. The authors observe changes in actin and myosin distribution during siphon tube morphogenesis using fixed specimens and immunohistochemistry. They discover that there is a biphasic change in the actomyosin localization that correlates with changes in cell shapes. Initially, there is the well-known relocation of actomyosin from the lateral sides to the apical surface of cells that will invaginate, accompanied by a concomitant lengthening of the central cells within the invagination, but not a lot of invagination. Coincident with a second, more rapid, phase of invagination, the authors see a relocalization of actomyosin back to the lateral sides of the cells. This 2nd "bidirectional" relocation of actin appears to be important because optogenetic inhibition of myosin in the lateral domain after the initial invaginations phase resulted in a block of further invagination. Although not noted in the paper, that the second phase of siphon invagination is dependent on actomyosin is interesting and important because it has been shown that during *Drosophila* mesoderm invagination that a second "folding" phase of invagination is independent of actomyosin contraction (Guo et al. *elife* 2022), so there appear to be important differences between the *Drosophila* mesoderm system and the ascidian siphon tube systems.*

Using the experimental data, the authors create a vertex model of the invagination, and simulations reveal a coupled mechanism of apicobasal tension imbalance and lateral contraction that creates the invagination. The resultant model appears to recapitulate many aspects of the observed cell behaviors, although there are some caveats to consider (described below).

We sincerely thank you for this insightful comment and for bringing the important study by Guo et al. (2022) to our attention. We fully agree that a direct comparison between these two mechanisms is important of our findings. As you astutely point out, the fundamental difference lies in the autonomy and driving force of the second, rapid invagination phase. To highlight this important conceptual advance, we will add a dedicated paragraph in the Discussion section to explicitly discuss this point.

Strengths:

The studies and presented results are well done and provide important insights into the physical forces of epithelial invagination, which is important because invaginations are how a large fraction of organs in multicellular organisms are formed.

Thank you for this positive assessment and for recognizing the significance of our work in elucidating the physical mechanisms underlying fundamental morphogenetic processes. We have striven to provide a comprehensive and rigorous analysis, and are grateful for this encouraging feedback.

Weaknesses:

(1) This reviewer has concerns about two aspects of the computational model. First, the model in Figure 5D shows a simulation of a flat epithelial sheet creating an invagination.

However, the actual invagination is occurring in a small embryo that has significant curvature, such that nine or so cells occupy a 90-degree arc of the 360-degree circle that defines the embryo's cross-section (e.g., see Figure 1A). This curvature could have important effects on cell behavior.

Thank you for bringing up the issue of tissue curvature. In this initial version of the model, we treated the tissue as flat because although the anterior epidermis indeed has significant curvature, the region that actually undergoes invagination occupies only a small arc of the embryo's cross-section—roughly 30-degree arc of the 360-degree circle. In addition, the embryo elongates anisotropically, and by 16.5 hpf the curvature has largely diminished (Fig.1A), leaving this local region effectively flattened. We agree that this simplification may overlook contributions from early curvature, and we will examine curvature changes more carefully in the data and incorporate curved geometry into the model to evaluate their impact.

(2) The second concern about the model is that Figure 5 D shows the vertex model developing significant "puckering" (bulging) surrounding the invagination. Such "puckering" is not seen in the in vivo invagination (Figure 1A, 2A). This issue is not discussed in the text, so it is unclear how big an issue this is for the developed model, but the model does not recapitulate all aspects of the siphon invagination system.

Thank you for pointing out the issue regarding the accuracy of the deformation pattern in our simulations. We do observe a mild puckering in vivo around 17 hpf (Fig. 1A), but it is clearly less pronounced than in the current model. The presence of such deformation suggests that bending stiffness of the epithelial sheet contributes to the mechanics of the invagination, which is included in our current model. While the discrepancy reflects limitations in our mechanical assumptions and geometric simplifications, including oversimplified interactions between the apical cell layer and the underlying basal cells, as well as the omission of tissue curvature. We will refine these aspects in the revised model to better reproduce the deformation patterns observed in vivo.

(3) In Figure 2A, Top View, and the schematic in Figure 2C, the developing invagination is surrounded by a ring of aligned cell edges characteristic of a "purse string" type actomyosin cable that would create pressure on the invaginating cells, which has been documented in multiple systems. Notably, the schematic in Figure 2C shows myosin II localizing to aligned "purse string" edges, suggesting the purse string is actively compressing the more central cells. If the purse string consistently appears during siphon invagination, a complete understanding of siphon invagination will require understanding the contributions of the purse string to the invagination process.

Thank you for this excellent observation. We agree that the ring-like actomyosin structure is a prominent feature during the initial stages of invagination, and its potential role warrants discussion. We carefully re-examined our data. Our analysis confirms that this myosin ring is most pronounced during the early initial invagination stage (approximately 13-14 hpf). This inward compression from the periphery would work in concert with apical constriction to help shape the initial invagination. However, this ring-like myosin pattern significantly diminishes in the accelerated invagination stage. We feel that the purse string may play a collaborative role in the early phase, however, its dissolution at the accelerated invagination stage indicates that Ciona atrial siphon invagination does not entirely rely on the sustained compression from the purse string of surrounding cells. These data will be included in the supplementary materials.

(4) The introduction and discussion put the work in the context of work on physical forces in invagination, but there is not much discussion of how the modeling fits into the

literature.

We apologize for not providing sufficient context on how our theoretical framework relates to prior work on the mechanics of invagination. You are absolutely right that the Introduction and Discussion sessions should more clearly situate our model within the existing literature, including the classical formulations it builds upon and the more recent models that address similar morphogenetic processes. In the revision, we will expand this section to acknowledge relevant work, clarify how our approach connects to and differs from previous models, and explicitly discuss the strengths and limitations of our framework. We appreciate this helpful suggestion and will make these connections much clearer.

Reviewer #2 (Public review):

Summary:

The authors propose that bidirectional translocation of actomyosin drives tissue invagination in Ciona siphon tube formation. They suggest a two-stage model where actomyosin first accumulates apically to drive a slow initial invagination, followed by translocation to lateral domains to accelerate the invagination process through cell shortening. They have shown that actomyosin activity is important for invagination - modulation of myosin activity through expression of myosin mutants altered the timing and speed of invagination; furthermore, optogenetic inhibition of myosin during the transition of the slow and fast stages disrupted invagination. The authors further developed a vertex model to validate the relationship between contractile force distribution and epithelial invagination.

Thank you for your thoughtful and accurate summary of our work and for your constructive critique.

Strengths:

- (1) The authors employed various techniques to address the research question, including optogenetics, the use of MRLC mutants, and vertex modelling.*
- (2) The authors provide quantitative analyses for a substantial portion of their imaging data, including cell and tissue geometry parameters as well as actin and myosin distributions. The sample sizes used in these analyses appear appropriate.*
- (3) The authors combined experimental measurements with computer modeling to test the proposed mechanical models, which represents a strength of the study. It provides a framework to explore the mechanical principles underlying the observed morphogenesis.*

We are grateful for your positive assessment of the multidisciplinary approaches, quantitative analyses, and the integration of modeling with experiments.

Weaknesses:

- (1) The concept of coordinated and sequential action of apical and lateral actomyosin in support of epithelial folding has been documented through a combination of experimental and modeling approaches in other contexts, such as ascidian endoderm invagination (PMID: 20691592) and gastrulation in Drosophila (PMIDs: 21127270, 22511944, 31273212). While the manuscript addresses an important question, related findings have been reported in these previous studies. This overlap reduces the degree of novelty, and it remains to be clarified how their work advances beyond these prior contributions.*

We thank you for raising this important point regarding the novelty of our work and for directing us to the key literature on ascidian endoderm invagination (PMID: 20691592) and *Drosophila* gastrulation (PMIDs: 21127270, 22511944, 31273212). We agree with the reviewer that the sequential activation of contractility in different cellular domains is a fundamental mechanism driving epithelial morphogenesis, as elegantly demonstrated in these prior studies. Our work builds upon this foundational concept. However, we believe we reveals a novel and distinct mechanical model: The ascidian endoderm and the atrial siphon involve a sequential shift of actomyosin contractility. However, the spatial pattern and functional outcomes are fundamentally different. In the ascidian endoderm (PMID: 20691592), the transition is from apical constriction to basolateral contraction. Basolateral contraction works in concert with a persistent circumferential to overcome tissue resistance and drive invagination. In contrast, our study of the atrial siphon reveals a bidirectional actomyosin redistribution between the apical and lateral domains. The basal domain in our system appears to play a more passive, structural role. While, *Drosophila* gastrulation also involves apical and lateral myosin, the mechanisms and dependencies differ. As supported by recent work (Guo et al. *elife* 2022), ventral furrow invagination can proceed even when lateral contractility is compromised, indicating that it is not an absolute requirement. In our system, however, optogenetic inhibition and our vertex model strongly suggest that the acquisition of lateral contractility is essential for the accelerated invagination stage. We will revise the text to better articulate these points of distinction and novelty in the Introduction and Discussion sections.

(2) One of the central statements made by the authors is that the translocation of actomyosin between the apical and lateral domains mediates invagination. The use of the term "translocation" infers that the same actomyosin structures physically move from one location to another location, which is not demonstrated by the data. Given the time scale of the process (several hours), it is also possible that the observed spatiotemporal patterns of actomyosin intensity result from sequential activation/assembly and inactivation/disassembly at specific locations on the cell cortex, rather than from the physical translocation of actomyosin structures over time.

Your critique regarding the term "translocation" was well-founded. We will replace "translocation" with the more accurate and conservative term "redistribution" throughout the manuscript, including in the title. We will also revise the text in the Results and Discussion sections to avoid overinterpretation.

(3) Some aspects of the data on actomyosin localization require further clarification. (1) The authors state that actomyosin translocation is bidirectional, first moving from the lateral domain to the apical domain; however, the reduction of the lateral actomyosin at this step was not rigorously tested. (2) During the slow invagination stage, it is unclear whether myosin consistently localizes to the apical cell-cell borders or instead relocates to the medioapical domain, as suggested by the schematic illustration presented in Figure 2C. (3) It is unclear how many cells along the axis orthogonal to the furrow accumulate apical and lateral myosin.

Thank you for your insightful comments, which will help us significantly improve the clarity and rigor of our actomyosin localization analysis. To address the points raised, we will undertake several key revisions: First, we will add new quantitative analyses of active myosin intensity from earlier time points (13-14 hpf) to rigorously support the initial lateral-to-apical redistribution phase. Second, we will correct the schematic in Figure 2C to accurately reflect the predominant localization of active myosin at the apical cell-cell borders. Finally, we will clarify that the actomyosin redistribution occurs within a broader domain of

approximately 15-20 cells in the invagination primordium, not being restricted to the single central cell on which our quantitative measurements were focused.

(4) The overexpression of MRLC mutants appears to be rather patchy in some cases (e.g., in Figure 3A, 17.0 hpf, only cells located at the right side of the furrow appeared to express MRLC T18ES19E). It is unclear how such patchy expression would impact the phenotype.

Thank you for your observation. We acknowledge that mosaic expression is common in Ciona electroporation. For all quantitative analyses, we only selected embryos in which the central cell, along with more than half of the surrounding cells in the primordium, showed clear expression of the plasmid.

(5) In the optogenetic experiment, it appears that after one hour of light stimulation, the apical side of the tissue underwent relaxation (comparing 17 hpf and 16 hpf in Figure 4B). It is therefore unclear whether the observed defect in invagination is due to apical relaxation or lack of lateral contractility, or both. Therefore, the phenotype is not sufficient to support the authors' statement that "redistribution of myosin contractility from the apical to lateral regions is essential for the development of invagination".

We agree that our optogenetic inhibition experiment does not distinguish between apical and lateral roles. To directly address this point, we will perform additional experiments in which we conduct the optogenetic inhibition and subsequently fix and stain the embryos for active myosin and F-actin. This will allow us to quantitatively compare the distribution of actomyosin in the light-stimulated experimental group versus the dark control group. We expect that light activation will have a more pronounced inhibitory effect on the lateral domains than on the apical domain, as the latter is naturally undergoing a reduction in contractility at this stage.

(6) The vertex model is designed to explore how apical and lateral tensions contribute to distinct morphological outcomes. While the authors raise several interesting predictions, these are not further tested, making it unclear to what extent the model provides new insights that can be validated experimentally. In addition, modeling the epithelium as a flat sheet and not accounting for cell curvature is a simplification that may limit the model's accuracy. Finally, the model does not fully recapitulate the deeply invaginated furrow configuration as observed in a real embryo (comparing 18 hpf in Figure 5D and 18 hpf in Figure 1A) and does not fully capture certain mutant phenotypes (comparing 18 hpf in Figure 5F and 18 hpf in Figure 3B right panel).

Thank you for raising these important points. We agree that several model predictions require stronger experimental grounding, and that the flat-sheet assumption is an oversimplification that likely contributes to the model not fully capturing certain morphological features. Our current simulations of myosin perturbation are largely consistent with the optogenetic experiments and the behavior of the myosin mutant. However, the predictions obtained by theoretically decoupling apical and lateral tension are difficult to validate experimentally, given the challenges of selectively manipulating these two components in vivo. Based on your helpful suggestions, we will extend the model to incorporate tissue curvature and examine how initial bending influences the mechanics of invagination, which we expect will improve the accuracy of the model's morphological predictions.

Reviewer #3 (Public review):

Summary:

In this manuscript by Qiao et al., the authors seek to uncover force and contractility dynamics that drive tissue morphogenesis, using the Ciona atrial siphon primordium as a model. Specifically, the authors perform a detailed examination of epithelial folding dynamics. Generally, the authors' claims were supported by their data, and the conceptual advances may have broader implications for other epithelial morphogenesis processes in other systems.

Thank you for your positive summary and for recognizing the broader implications of our work.

Strengths:

The strengths of this manuscript include the variety of experimental and theoretical methods, including generally rigorous imaging and quantitative analyses of actomyosin dynamics during this epithelial folding process, and the derivation of a mathematical model based on their empirical data, which they perturb in order to gain novel insights into the process of epithelial morphogenesis.

Thank you for highlighting the strengths of our multidisciplinary methodology.

Weaknesses:

There are concerns related to wording and interpretations of results, as well as some missing descriptions and details regarding experimental methods.

We will revise the manuscript to address your concerns regarding wording and methodological details. Your feedback led us to improve clarity, precision, and the depth of methodological description throughout the text.

<https://doi.org/10.7554/eLife.108588.1.sa0>